

Climate change beliefs and savings behavior: a macroeconomic perspective

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What are the macro implications of climate change beliefs via savings?

1. Do beliefs over climate change affect individual savings?
2. Does the effect on capital accumulation have macroeconomic consequences?

Do climate beliefs affect individual savings?

Do climate beliefs matter for macroeconomic outcomes via capital accumulation?

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- Analytical model to uncover trade-offs and rationalize findings

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Do climate beliefs matter for macroeconomic outcomes via capital accumulation?

- **Non-stationary** general equilibrium model with incomplete markets and aggregate risk
 - Climate change as shift in stochastic productivity process
 - Uncertainty over trend in productivity, but learning from realizations
 - Heterogeneity in income, wealth and beliefs

Key take-aways: climate change beliefs matter.

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...and affect macroeconomic outcomes during the climate transition.

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- **Climate concern effect:** capital $\uparrow$ output loss from climate change $\downarrow$
→ NPV of output losses: +12% under too optimistic versus accurate beliefs
- Increase in capital benefits the poor disproportionately: wealth inequality $\downarrow$
→ Gini: -0.2% during transition relative to initial value
→ Without savings adjustment: Gini $\uparrow$ during transition
- Small but persistent impacts of disagreement:
→ NPV of output losses: +1.5% under dispersed relative to homogenous beliefs

Climate change impacts on the macroeconomy

- Stochastic damages: Golosov et al. (2014), Cai and Lontzek (2019)
- Anticipation: Bilal and Rossi-Hansberg (2023), Bakkensen and Barrage (2022)

Contribution: • Amplified consequences of climate uncertainty in decentralized framework

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Incomplete markets and aggregate risks: theory and computation

- Krusell and Smith (1998), Farhi et al. (2022), Broer et al. (2022), Fernández-Villaverde et al. (2023), Bilal (2023), Auclert et al. (2021), Moll (2024)

Contribution: • Non-stationarity in aggregate process
 • Isolating role of beliefs in stochastic framework
 • Global solution method

Climate beliefs and individual savings:
evidence and theory

Dynamic general equilibrium model

Macroeconomic effects of climate beliefs

Climate beliefs and individual savings: evidence and theory

Do climate beliefs affect savings choices?

Empirical evidence:

1. Observational evidence: UK Understanding Society Survey, 2 waves
 - Climate concern index, savings choices, demographics
 - Significant correlation between CC concern and savings on intensive and extensive margin
Likelihood to save +9pp, savings share +1pp
2. Causal evidence: Specific purpose online survey

Goal: obtain causal evidence by using within-subject design (cf. Andre et al., 2025)

Hypothetical savings choice: How much would you save out of a 5000 GBP transfer?

- For each of 2 different climate impact scenarios: Increasing $\mathcal{I}$, Constant $\mathcal{C}$
- Presented in random order

Key statistic: relative effect of scenario on marginal savings $\Delta s = \frac{\text{Saved}|\mathcal{I} - \text{Saved}|\mathcal{C}}{\text{Saved}|\mathcal{C}}$

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	Full sample
Δs	7.6%***
Observations	543
average MPS $\mathcal{C}$	68.2%

low	medium	high
10.0%***	7.2%**	5.3%
178	230	81
66.5%	67.3%	73.3%

by income level

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Theory: Consumption-savings model in partial equilibrium

- First-order response to perturbation in probabilities of future states from Euler equation
- Key drivers: variation in marginal value of wealth across states, marginal propensity to save
- Rationalizes positive, heterogeneous savings response

Key take-aways:

- Empirical evidence: positive relationship between climate concern and savings choices
- Passthrough of adverse aggregate risks: savings $\uparrow$ depending on idiosyncratic state

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Open question: Do climate change beliefs matter for the macroeconomy?

- How does additional capital accumulation affect output losses during **climate transition**?
- Does the **observed disagreement** over climate change matter for **aggregate outcomes**?

→ Dynamic general equilibrium model

Dynamic general equilibrium model

Atomic households

- are subject to exogenous idiosyncratic shocks (skills, life-cycle);
- consume and save in assets to maximize expected utility given budget.

Representative firm

- produces using capital and labor, pays wages and interest according to profit maximization
- is subject to multiplicative productivity shocks $\zeta_t \in \{\zeta^L, \zeta^H\}$, $\zeta^L < \zeta^H$

Market clearing: Capital supply equals households' savings.

Climate change

- causes exogenous and deterministic increase in global temperature T_t ;
- which is known at time $t = 0$ and may affect the frequency of low productivity states.
→ Non-stationarity

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Probability of adverse aggregate shock is given by $\mathbb{P}(\zeta_t = \zeta^L) = p^{1-\gamma T_t}$ with **unknown** parameter γ

- Two possible states of the world: $\gamma \in \{0, \bar{\gamma}\}$
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- Bayesian updating

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Impacts modeled via **probability**, levels stay constant

- **Any** draw $\{\zeta_t\}_t$ possible under **all** beliefs
- Implicit level effect

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→ distribution over households Ψ is infinite dimensional state variable

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→ distribution over households Ψ is infinite dimensional state variable

Assumption: bounded rationality and dynamic equilibrium

- Perceived Law of Motion (PLM)

$$K' = \mathcal{H}(K; \mathcal{X}) \quad \text{for some explanatory variables } \mathcal{X}.$$

- For given PLM $\mathcal{H}$, initial values and a shock sequence, the *dynamic equilibrium* of the economy is a sequence of distributions $\{\Psi_t\}_{t \geq 0}$ so that:
 1. Each period, households and firms optimize given their beliefs; markets clear.
 2. Ψ_t evolves according to (i) demographic dynamics, (ii) savings choices, and (iii) Bayes' formula.

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Baseline choice of PLM:

find self-justified equilibrium to approximate rational expectations (Krusell and Smith, 1998)

Earnings, life-cycle and preferences:

- AR(1) process for skills estimated on UK panel data
- Expected duration working life (retirement): 40 (20) years; replacement rate 60%
- CRRA utility with $\sigma = EIS^{-1} = 2$, β to match $K/Y = 2.7$ before transition

Aggregate shocks: Effect of climate change on UK productivity

- AR(1) process for temperature, long-run increase of 0.9 deg within 75 years.
- Meta study Rising et al. (2022) estimates output losses:

1.1% in 2022, 3.3% by 2050.

- Bad states $p_0 = 0.15$, $\zeta^L = 0.93$:

$$\mathbb{E}(\zeta_0) = 0.9895, \mathbb{E}(\zeta_{100}) = 0.965 \text{ with } p_{100} = 0.5, \bar{\gamma} = \frac{1}{0.9} \left(1 - \frac{\ln 0.5}{\ln p_0} \right).$$

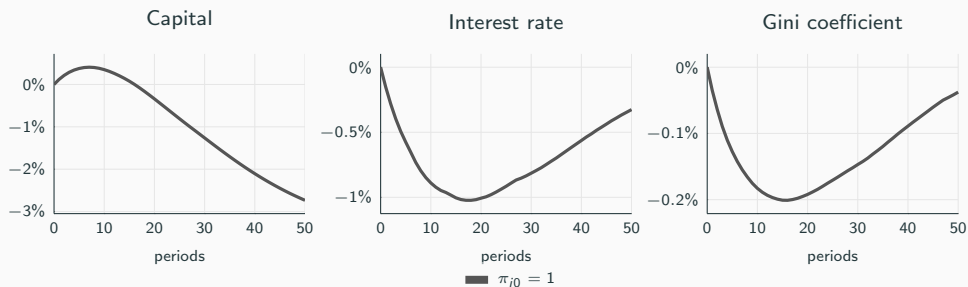
High damages counterfactual: γ^H s.t. $p_{100} = 0.8$, $\mathbb{E}(\zeta_{100}) = 0.944$.

Macroeconomic effects of climate beliefs

The climate transition

Baseline: $\bar{\gamma} = \gamma$

How does the climate transition affect the macroeconomy if **everyone expects** productivity impacts?

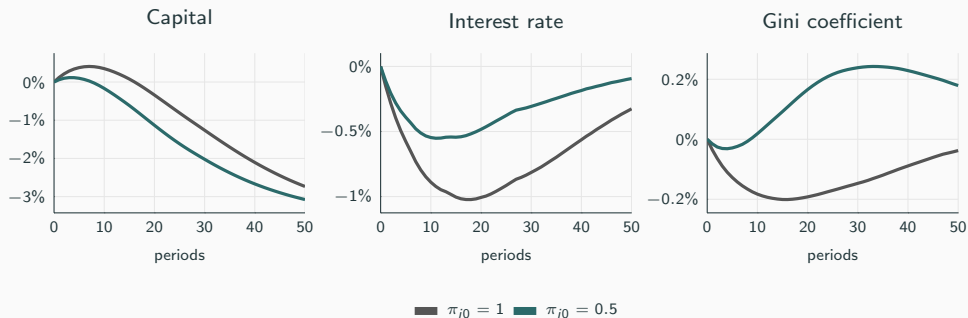


Ensemble averages over time, relative to initial SSS

The role of beliefs during the climate transition

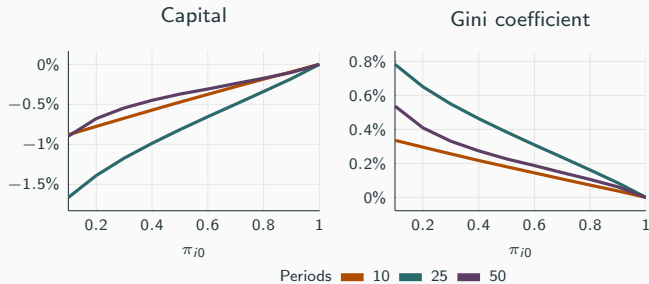
Capital ↓ if population assumes 50% chance of no productivity impacts

→ wealth inequality rises



Ensemble averages over time, relative to initial SSS

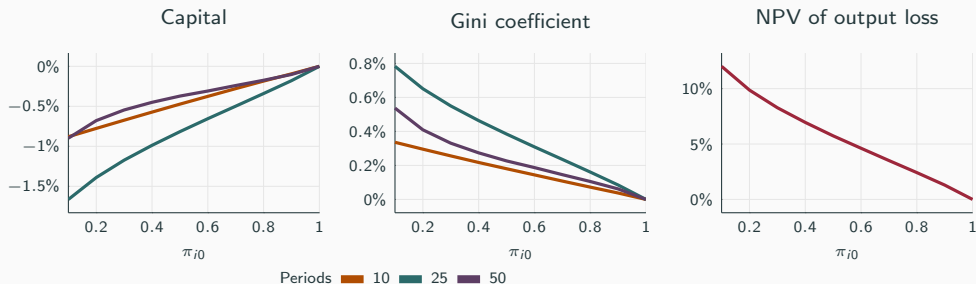
The role of beliefs during the climate transition



Relative deviation to $\pi_{i0} = 1$, no belief heterogeneity

Divergence for $\pi = 0$

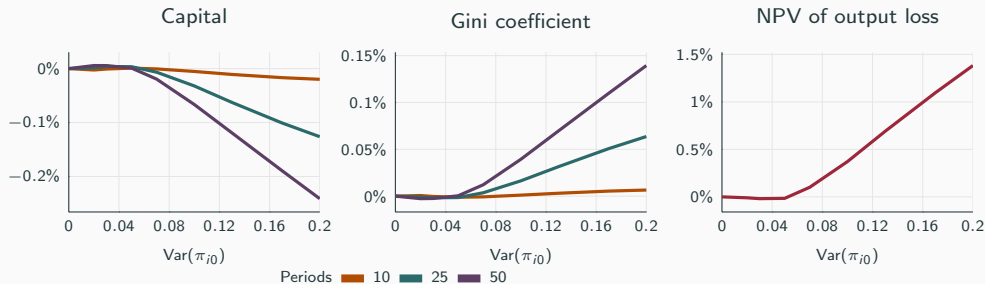
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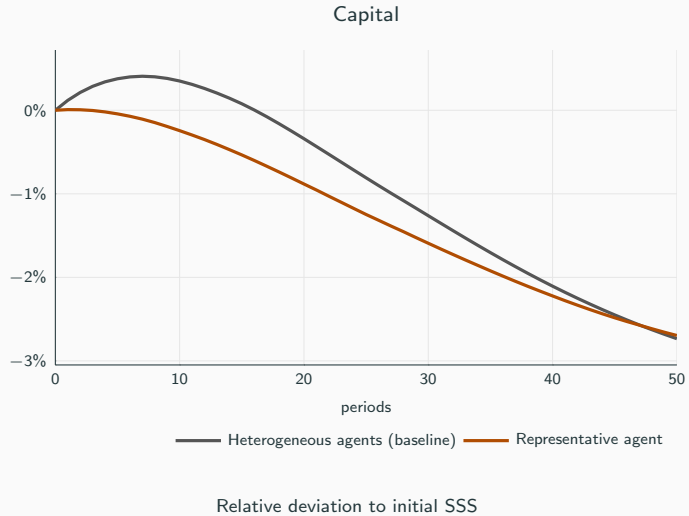
The effects of belief dispersion



Relative deviation to $\text{Var}(\pi_{i0}) = 0$ at $\mathbb{E}(\pi_{i0}) = 0.7$.

- Capital $\downarrow$ Gini $\uparrow$ under mean-preserving spread
- Persistent: deviation rises even after 25 years
- Small effects compared to variation in first moment

The role of incomplete markets



Individual behavior

- Climate concern increases individual savings in model of aggregate climate damages
- Consistent with UK survey evidence

Aggregate and distributional effects

- Climate concern effect: capital increases in short-run, attenuates climate damages
- Heterogeneous impacts: wealth inequality ↓
- Disagreement: small but persistent role

Beliefs over aggregate risks in incomplete markets framework

- Development and solution of a non-stationary model
- Broadly applicable methodology: uncertainty and disagreement over structural change

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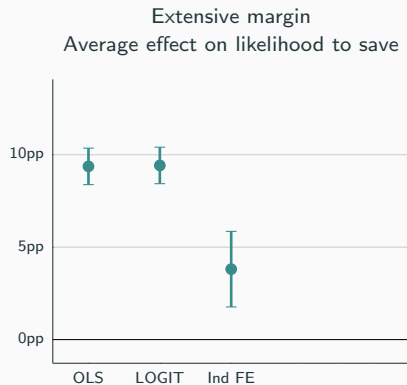
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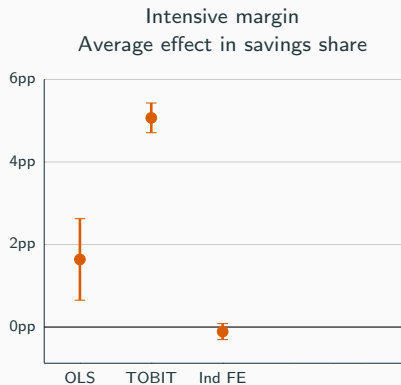
Appendix

- Climate change deniers $\pi_{i0} = 0$
→ endogenous polarization
- Overestimation of climate damages
→ asymmetric effects of inaccurate beliefs
- Alternative PLM
→ memory matters

Empirical results: full versus no concern

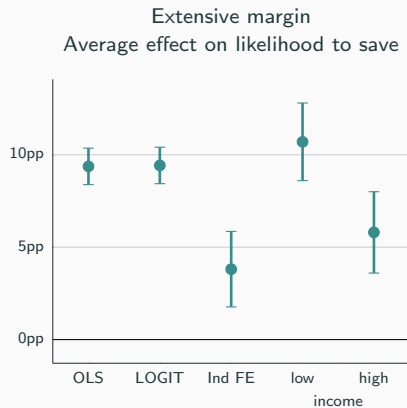


Share of savers: 46.5%

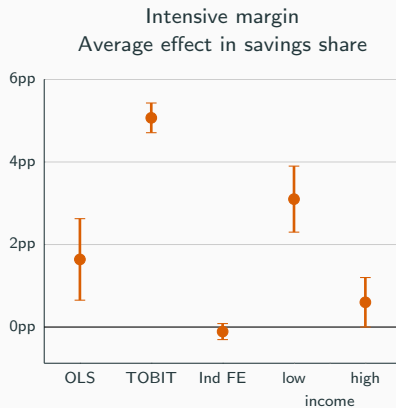


Average savings share: 6.45%

Empirical results: full versus no concern



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Average savings share: 6.45%

Savings response to a change in climate change beliefs

Standard consumption-savings problem: an individual i

- consumes c_{it} and saves $a_{i,t+1}$ in assets to maximize expected utility given budget

$$\max_{c_{it}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}) \right]$$
$$s.t. \ c_{it} + a_{i,t+1} = R_t a_{it} + y_{it}, \ a_{i,t} \geq 0 \ \forall t';$$

- faces uncertainty over future returns r_t and income y_{it} .
 - Aggregate productivity: average wages and asset returns
→ affected by climate change
 - Idiosyncratic state: labor income
→ independent of climate change

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How do savings depend on climate beliefs?

- First order response to perturbation in probabilities of future states from Euler equation (cf. Farhi et al., 2022)

An increase in $p' = \mathbb{P}(\text{low productivity next period})$ affects consumption choice today:

$$\frac{d \ln(c)}{dp'} = -\varepsilon(c) \cdot MPS \cdot \mathcal{D}.$$

- $\mathcal{D}$: Relative effect of low draw on expected marginal value of wealth
 - Under CRRA utility with $EIS < 1$ and standard parametrization $\mathcal{D} \geq 0$
 - Amplified by incomplete markets
- MPS : Marginal propensity to save
- $\varepsilon(c)$: Elasticity of intertemporal substitution

Take-aways:

1. Concern over adverse macro risks: individual savings $\uparrow$ in standard model
2. Opposing forces of current idiosyncratic state: large effect for **poor savers**

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- Future impacts: history dependence, risk-adjusted measure

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First order consumption response

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 - Under CRRA utility with $EIS < 1$ and standard parametrization $\mathcal{D} \geq 0$
 - Amplified by incomplete markets
 - High for **low asset holding, low expected income**
- MPS : Marginal propensity to save
 - Small if **borrowing constrained**
- $\varepsilon(c)$: Elasticity of intertemporal substitution
- Future impacts: history dependence, risk-adjusted measure

Take-aways:

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2. Opposing forces of current idiosyncratic state: large effect for **poor savers**

Consider a hypothetical situation in which you unexpectedly receive a one-time payment of 5000 GBP today. This payment is unconditional. We want to know how much of this money you would save, taking a specific scenario for granted. Any money which you do not save is assumed to be spent. Below, we provide a brief description of what we mean by saving and spending.

Saving means that, instead of using money today, you reserve it for future use. Examples of savings include cash reserves, money in bank accounts, retirement accounts, financial assets or real estate. Repaying debt is also an important form of saving. By repaying debt today, you owe less money in the future, which means that more money is available for future use.

Spending includes all money spent on goods and services, including rent. Goods include durable goods (such as electronics, furniture, or car maintenance) and nondurable goods (such as groceries, vacations, or gasoline).

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Scenario A: Assume that currently, climate-related events with significant negative consequences to the UK happen on average once every 7 years, and that the average frequency **increases** due to climate change. Assume that these events will happen, on average, twice as often by 2040 (once every 3.5 years) and more than three times as often by 2100 (once every other year).

Scenario B: Assume that currently, climate-related events with significant negative consequences to the UK happen on average once every 7 years, and that the average frequency **will not change until 2100**.

Participants use a slider to indicate how much of the hypothetical one-time payment of 5000 GBP they would choose to save under each scenario. They can choose increments of 250 GBP.

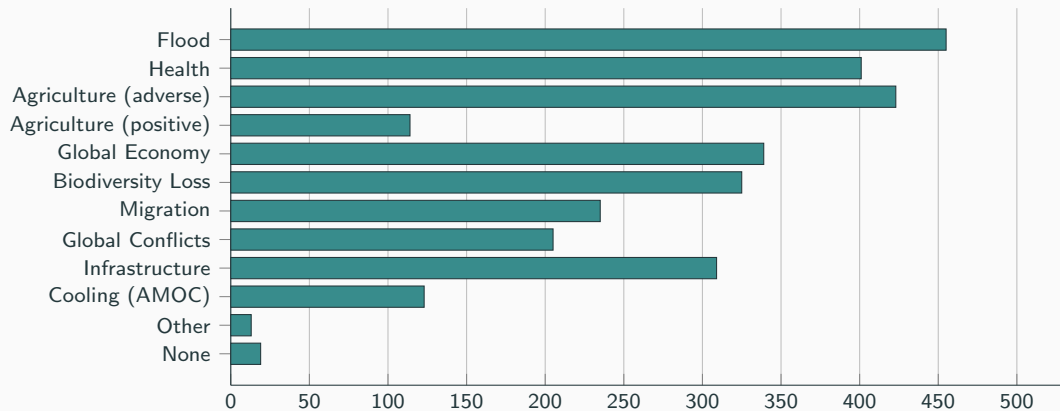
The questions are shown in random order, and there is no option to move back to the first question.

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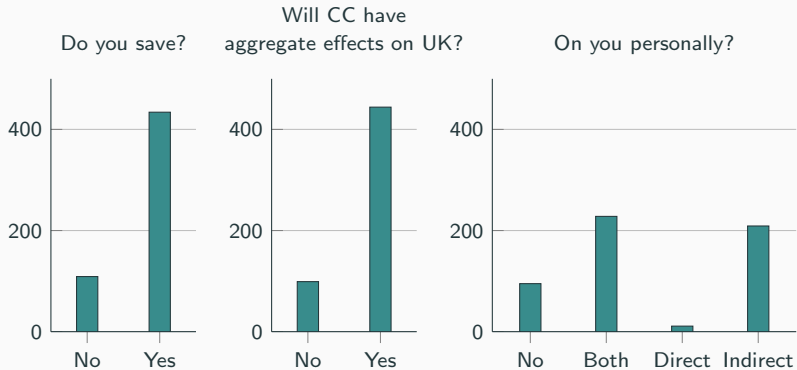
- conducted in October 2025
- pilot study (20 participants) for power analysis
- target: 600 participants; final sample: 543
- Recruitment via prolific
- Sample criterion: UK resident, older than 18;
- additional data: age, individual and household income bracket, nationality, ethnicity, children.

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Mechanisms



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Two types of uncertainty:

- Idiosyncratic $\phi \in \Phi$: skills and demographics
- Aggregate $\zeta \in Z = \{\zeta^L, \zeta^H\}$: productivity

Agent i chooses consumption to maximize expected utility over life-time

$$\max_{c_{it}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}) \right].$$

under some probability distribution, subject to budget constraint

$$c_{it} + a_{it+1} = y(\zeta_t, \phi_{it}) + R(\zeta_t)a_{it}, \quad a_{it+1} \geq 0.$$

For multiplicative productivity shocks: $R(\zeta^L) < R(\zeta^H)$, $y(\zeta^L, \cdot) < y(\zeta^H, \cdot)$, but

$$y(\zeta^L, \cdot)/y(\zeta^H, \cdot) < R(\zeta^L)/R(\zeta^H).$$

Let $\theta^t = (\theta_0, \dots, \theta_t)$ denote the history of shocks up to time t . A perturbation in future probabilities $p_t =$ then leads to the following first order response:

$$\frac{dc_0}{c_0} = -\varepsilon_0 \sum_{t=0}^{\infty} \left(\sum_{\theta^t} \mathbb{P}^*(\theta^t) \left(\prod_{j=0}^t MPS(\theta^j) \right) \mathcal{D}_t(\theta^t) \right) dp_{t+1}. \quad (1)$$

where $\mathcal{D}_t$ is the relative effect of low draw on expected marginal value of wealth

$$\mathcal{D}_t = \frac{\mathbb{E}[V'(a_{t+1}, \theta^{t+1}) | \zeta_{t+1} = \zeta^L] - \mathbb{E}[V'(a_{t+1}, \theta^{t+1}) | \zeta_{t+1} = \zeta^H]}{\mathbb{E}[V'(a_{t+1})]},$$

and $\mathbb{P}^*$ is the risk-adjusted probability measure over θ^t

$$\mathbb{P}^*(\theta^t | \theta^{t-1}) = \mathbb{P}(\theta^t | \theta^{t-1}) \frac{V'(a_t, \theta^t)}{\mathbb{E}[V'(a_t, \theta^t)]}.$$

back

Non-stationary process: distinguish between time averages and ensemble averages

→ use smart sample for shock matrix $\zeta = (\zeta_{it}) \in \mathbb{R}^{n_{sim} \times T_{sim}}$ to estimate system statistics

- In each period t , find $\tilde{n}_t^L = \lfloor p_t \cdot n_{sim} \rfloor$ as the closest integer to theoretically implied number of low aggregate draws and set

$$\hat{\zeta}^t = (\zeta^L \cdot \mathbf{1}_{\tilde{n}_t^L \times 1}) \times (\zeta^H \cdot \mathbf{1}_{n_{sim} - \tilde{n}_t^L \times 1}) \in \mathbb{R}^{n_{sim} \times 1}$$

- Randomly draw a permutation p on $\{1, \dots, n_{sim}\}$ and set

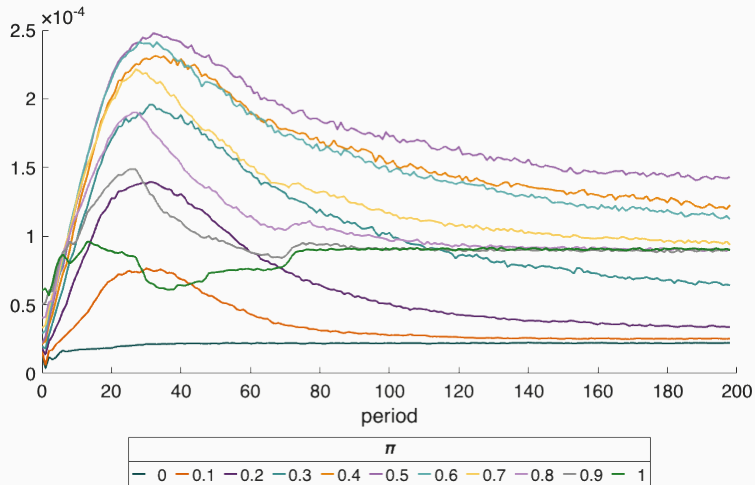
$$\zeta_{it} = \hat{\zeta}_{p(i)}^t.$$

→ match theoretical moments of ζ for each time t closely even for small sample sizes n_{sim}

Validation: matching moments of endogenous variables

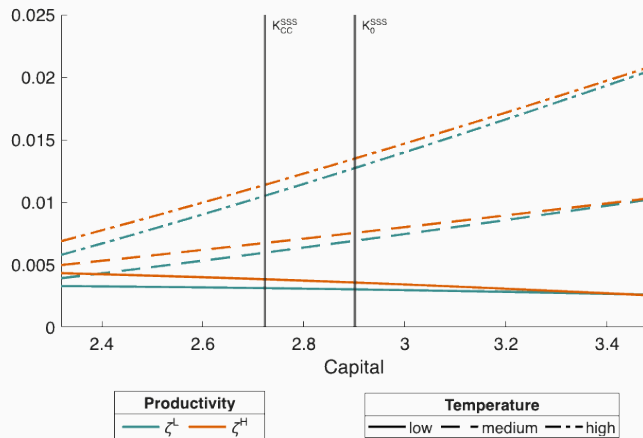
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PLM performance



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Heterogeneity in capital forecasting



Difference $\mathcal{H}_1 - \mathcal{H}_0$

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Adaptive expectations with memory

Fully persistent (static) beliefs easy: $\mathcal{H}(K) = K$

Updating $\mathcal{H}$ over time:

- Fix a draw $\{\zeta_t\}$, func. form $\mathcal{H}^{adp}(K, \zeta) = \mathcal{F}(K, \zeta; \Theta)$, initial coefficients Θ_0 , cov. matrix Σ_0
- Fix interval length $s > 0$, so that $\mathcal{H}_{t_s^*}^{adp} = \mathcal{H}_{t_s^*+i}^{adp}$ for all $i = 0 \dots s-1$. At any $t_s^* = t \cdot s$, update PLM given observations

$$\mathbf{Y}_s = (\log(K_{t+1}))_{t=t_s^*+i} \text{ and } \mathbf{X}_s = (1, \log(K_t), \mathbb{1}_{\zeta_t=\zeta^L}, \mathbb{1}_{\zeta_t=\zeta^L} \cdot \log(K))_{t=t_s^*+i}$$

1. Infinite memory

$$\Sigma_s = \mathbf{X}_s^\top \mathbf{X}_s + \Sigma_{s-1}$$

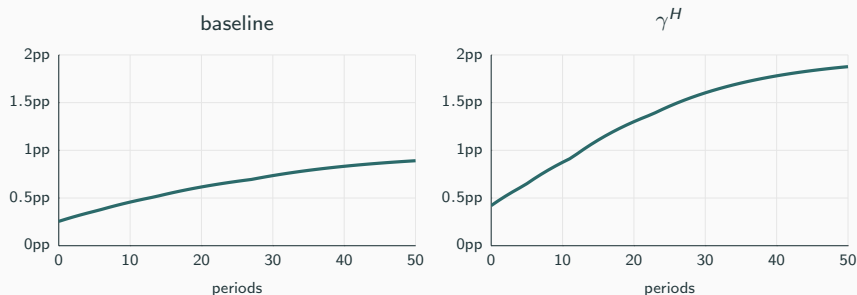
$$\Theta_s = \Theta_{s-1} + \Sigma_s^{-1} \mathbf{X}_s^\top [\mathbf{Y}_s - \mathbf{X}_s \Theta_{s-1}]$$

2. Finite memory: introduce parameter λ and find

$$\Theta_s = \lambda \Theta_{s-1} + (1 - \lambda)(\mathbf{X}_s^\top \mathbf{X}_s)^{-1} \mathbf{X}_s^\top \mathbf{Y}_s$$

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Average effects on individual savings choices

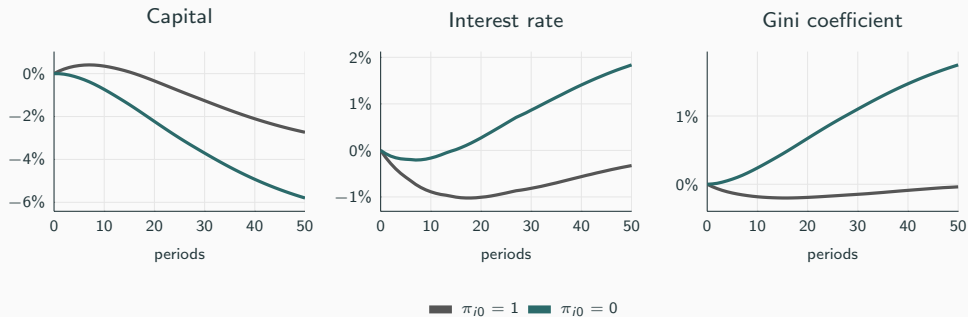


Difference in average savings share between $\pi = 1$ and $\pi = 0$
Under rising temperature, with constant capital

Changes in likelihood: 1.91pp vs 1.93pp initially

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The case of ignorance

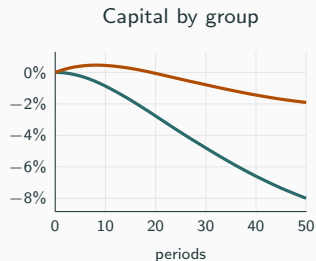
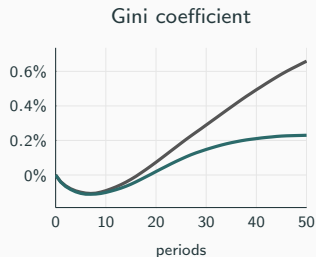
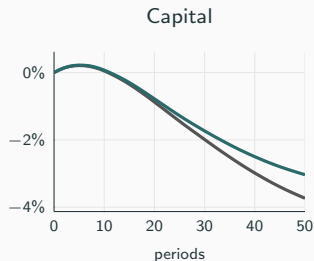


Ensemble averages over time, relative to initial SSS
Homogenous beliefs

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Polarization under extreme beliefs

$\mathbb{E}(\pi_{i0}) = 0.7$ under ignorance



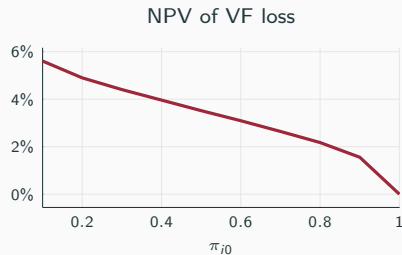
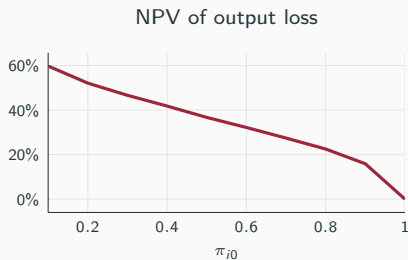
■ heterogeneous and extreme ■ homogenous beliefs

■ $\pi_{i0} = 0$ ■ $\pi_{i0} = 1$

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Overestimated climate damages

Individual beliefs γ^H vs reality γ

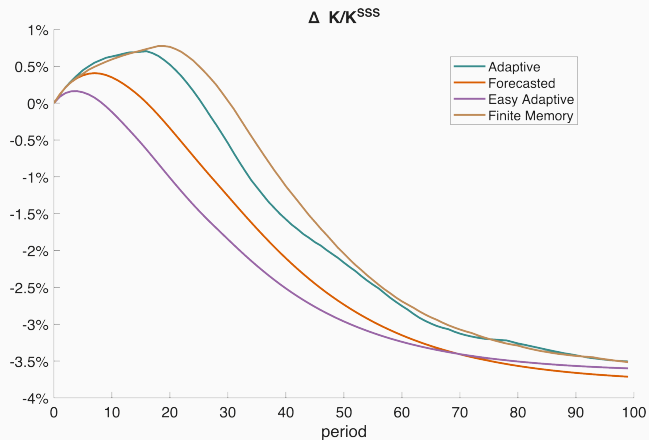


Relative to $\pi_0 = 1$

Welfare effect: gains for future generations

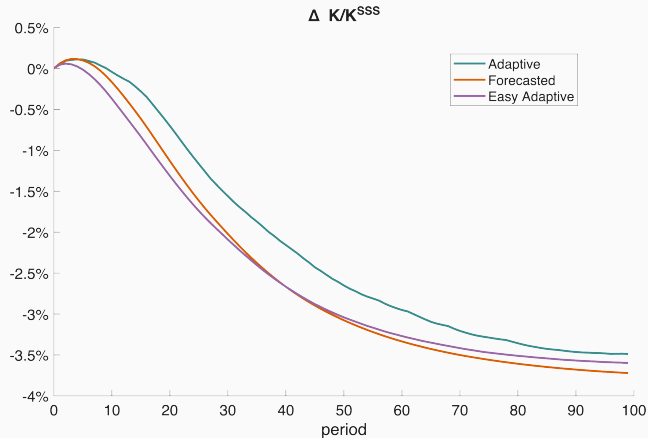
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Capital dynamics under accurate climate beliefs - PLM comparison



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Capital dynamics under $\mu = 0.5$ - PLM comparison



Computationally intensive: adaptive takes about 8 hours for 150 simulations

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